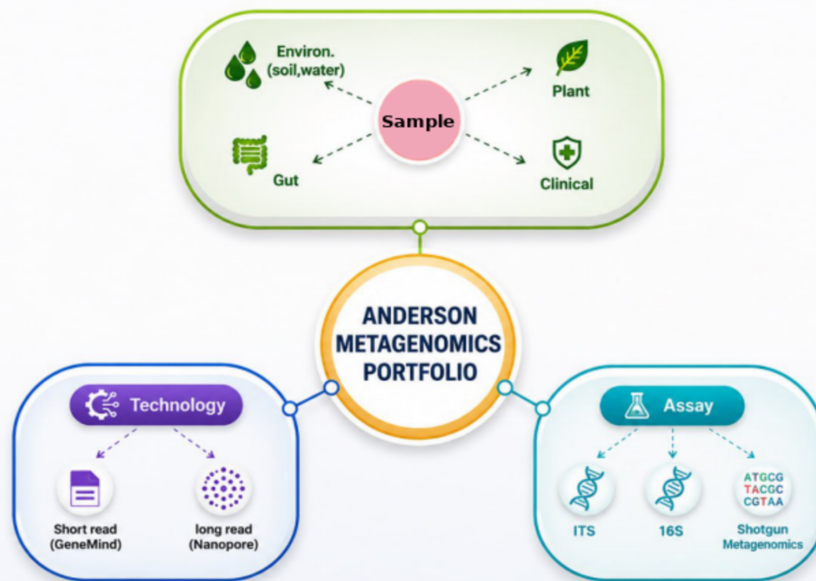


## METAGENOMICS SEQUENCING SERVICES

- Sequences the complete microbial community directly from clinical, agricultural, and environmental samples — no culture required.
- Simultaneous identification of bacteria, fungi, viruses, archaea, and protozoa from a single sequencing run.
- Integrated AMR gene profiling, functional pathway analysis, and strain-level resolution in one validated workflow.

**Anderson Metagenomics Panel** — Whole Metagenome Shotgun | 16S rRNA Community Profiling | ITS Mycobiome | Clinical Pathogen Detection (blood, CSF, BAL, stool) | Gut & Urobiome Analysis | Plant & Rhizosphere Profiling | Environmental Surveillance | Nanopore Long-Read Assembly | AMR Genotyping.



### Sample-to-Insight Workflow

6 stages · one validated pipeline



### METAGENOMICS PORTFOLIO

ASSAYS (1-3) · SEQUENCING TECHNOLOGIES (4-5) · APPLICATION PANELS (6-9)

#### ASSAYS

|                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>1 16S rRNA Sequencing</b><br/>BACTERIAL PROFILING</p> <p>Cost-effective bacterial community profiling via conserved ribosomal markers. High-throughput diversity analysis across stool, soil, and water. Ideal for large-cohort microbiome studies. Archaeal profiling may be supported with appropriate primer selection.</p> | <p><b>2 ITS Sequencing</b><br/>FUNGAL COMMUNITY PROFILING</p> <p>Species-level fungal identification via ITS1/ITS2 amplicons. High-resolution mycobiome profiling across clinical, agricultural, and environmental matrices.</p> | <p><b>3 Whole Metagenome Shotgun</b><br/>WHOLE-COMMUNITY DNA</p> <p>Culture-free sequencing of all microbial DNA. Simultaneous taxonomy, AMR profiling, and functional pathway analysis for comprehensive community characterisation.</p> <ul style="list-style-type: none"> <li>&gt; Multi-kingdom detection</li> <li>&gt; Functional pathway analysis</li> <li>&gt; AMR determinant profiling</li> <li>&gt; MAG reconstruction</li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### SEQUENCING TECHNOLOGIES

|                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>4 Short-Read Sequencing</b><br/>GeneMind PLATFORM</p> <p>High-accuracy, high-depth reads for robust species ID and functional annotation. Scalable and cost-effective for large cohort studies. Compatible with 16S, ITS, and whole-metagenome strategies.</p> | <p><b>5 Long-Read Sequencing</b><br/>NANOPORE PLATFORM</p> <p>Kilobase-scale reads enabling complete genome assembly and strain-resolved taxonomy. Real-time pathogen ID with adaptive sampling. Full AMR gene chromosomal context. Compatible with 16S rRNA, ITS amplicon, and WMS workflows.</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### APPLICATION PANELS

|                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>6 Clinical Metagenomics</b><br/>PATHOGEN DETECTION</p> <p>Culture-free multi-kingdom ID from blood, CSF, BAL, stool, and tissue. Concurrent AMR genotyping.</p>                                                                                       | <p><b>7 Gut &amp; Urobiome Analysis</b><br/>INTESTINAL MICROBIOTA</p> <p>Comprehensive intestinal and urinary profiling linking dysbiosis to IBD, metabolic syndrome, and UTI. Supports therapeutic monitoring and restoration.</p>           |
| <p><b>8 Plant &amp; Rhizosphere Profiling</b><br/>RHIZOSPHERE &amp; ENDOPHYTES</p> <p>Root-zone and internal plant microbiota profiling. Identifies growth-promoting bacteria, mycorrhizal fungi, and phytopathogens across rhizosphere and endophytes.</p> | <p><b>9 Environmental Surveillance</b><br/>SOIL · WATER · SEDIMENT</p> <p>Biodiversity assessment and AMR surveillance across terrestrial and aquatic matrices. Detects indicator organisms and novel pathways for regulatory compliance.</p> |

## Advantages & Trade-Offs

what to weigh before you sequence

### + Advantages

- ▶ **Culture-independent** — Direct profiling from any sample.
- ▶ **Multi-kingdom** — Bacteria, fungi, viruses, archaea, protozoa.
- ▶ **AMR detection** — Resistance genes identified genome-wide.
- ▶ **Functional profiling** — Metabolic pathways and virulence factors mapped.
- ▶ **Strain-resolved** — Species-to-strain resolution with long reads.
- ▶ **Scalable** — Single samples to large population cohorts.
- ▶ **Broad matrix compatibility** — Clinical and environmental.
- ▶ **Fast turnaround** — Actionable results in 5–10 business days.

### – Trade-Offs

- ▶ **Higher cost** — Shotgun exceeds amplicon-based pricing.
- ▶ **Host contamination** — Human reads dilute microbial signal.
- ▶ **Bioinformatics complexity** — Specialist expertise required.
- ▶ **Database limits** — Novel organisms may be missed.
- ▶ **Low biomass samples** — Sparse samples need enrichment.
- ▶ **Extraction bias** — Lysis protocols may skew community data.
- ▶ **Longer workflow** — Complex projects: 7–14 days report.
- ▶ **Expert interpretation** — Multi-kingdom data needs specialist review.

## Sample Type Requirements

13 accepted matrices · colour coded by category

| SAMPLE TYPE        | APPLICATION                        | WHAT IT ENABLES                                                                    |
|--------------------|------------------------------------|------------------------------------------------------------------------------------|
| Stool              | Gut microbiome · IBD · Oncology    | Full community profiling; dysbiosis mapping; therapeutic monitoring                |
| Blood              | Sepsis · Bloodstream infections    | Culture-free pathogen ID and AMR genotyping from primary material                  |
| Urine              | UTI · Urobiome research            | Low-biomass urobiome characterisation; fastidious uropathogen detection            |
| Sputum             | Pulmonary infections · Lung        | Respiratory pathogen detection; lung dysbiosis in CF and COPD                      |
| Skin Swab          | Dermatology · Wound infection      | Skin microbiome diversity; wound pathogen ID and resistance genotyping             |
| Tissue Biopsy      | Tumour microbiome · Infection      | Intratumoral microbial signatures; sterile-site infection source ID                |
| Endometrial Tissue | Gynaecology · Fertility            | Uterine microbiota; Lactobacillus dominance; infertility & endometriosis dysbiosis |
| Plant Tissue       | Endophyte profiling · Phytobiome   | Internal plant microbiota; beneficial vs. pathogenic colonisation                  |
| Rhizosphere Soil   | Root-zone microbiome · Agriculture | Root-associated community; growth-promoting organism profiling                     |
| Soil               | Ecosystem health · Soil fertility  | Bulk biodiversity; carbon cycling genes; contamination assessment                  |
| Water              | Drinking water safety · Ecology    | Pathogen surveillance; indicator organisms; aquatic community diversity            |
| Wastewater         | AMR surveillance · Epidemiology    | Population-level AMR gene tracking; pathogen prevalence monitoring                 |
| Environmental Swab | Surface contamination · Biosafety  | Facility microbiome mapping; contamination source tracing                          |

## Recommended Sequencing Depth

by organism group · typical starting points

|                                 |                                     |                                       |                                      |
|---------------------------------|-------------------------------------|---------------------------------------|--------------------------------------|
| <b>Bacteria</b><br>5–10 M reads | <b>Fungi</b><br>10–20 M reads       | <b>Archaea</b><br>5–10 M reads        | <b>Protozoa</b><br>20–40 M reads     |
| <b>Viruses</b><br>20–50 M reads | <b>Shotgun MGX</b><br>20–50 M reads | <b>Clinical MGX</b><br>30–100 M reads | <b>Long-Read MGX</b><br>5–20 Gb data |

## Short-Read vs Long-Read Platform

choose based on your study goal

| Short-Read · GeneMind                                         | Long-Read · Nanopore                                         |
|---------------------------------------------------------------|--------------------------------------------------------------|
| ✓ <b>Accuracy</b> >99.9% per base (Q30+)                      | ✓ <b>Read length</b> >1 kb–2 Mb · complete chromosomes       |
| ✓ <b>Cost-effective at depth</b> Lower cost per base          | ✓ <b>Strain-resolved</b> Full-length reads · high resolution |
| ✓ <b>Large cohorts</b> Scalable, high-throughput              | ✓ <b>AMR gene context</b> Full chromosomal context           |
| ✗ <b>Read length</b> 150–250 bp · fragmented contigs          | ✓ <b>Real-time output</b> Adaptive sampling · 48 hrs         |
| ✗ <b>Strain resolution</b> Moderate · limited SVs             | ✗ <b>Cost per base</b> Higher than short-read                |
| ✗ <b>AMR context</b> Known alleles only                       | ✗ <b>Accuracy</b> 95–99% per base (Q20+)                     |
| ★ <b>Best for</b> Population studies · high depth · diversity | ★ <b>Best for</b> Assembly · strain ID · AMR · novel orgs    |

## Detection Capabilities

what a single run identifies

|           |                   |                 |                     |                 |
|-----------|-------------------|-----------------|---------------------|-----------------|
| Bacteria  | Fungi             | Viruses         | Archaea             | Protozoa        |
| AMR Genes | Virulence Factors | Mobile Elements | Functional Pathways | Novel Organisms |

**i** Values are general guidance, not fixed cut-offs. Optimal sequencing depth, QC thresholds, and input requirements shift with matrix complexity and study design. Our team tailors every workflow — extraction, library strategy, and bioinformatics pipeline — to your project before sequencing begins.